

CCAST, NOAA, and UW three-way CrIS Calibration Algorithm Comparison

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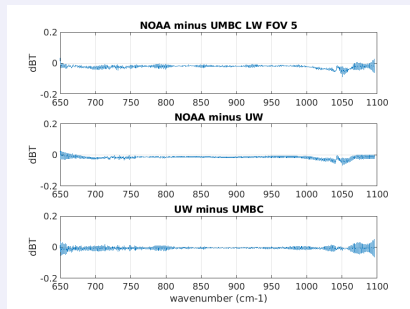
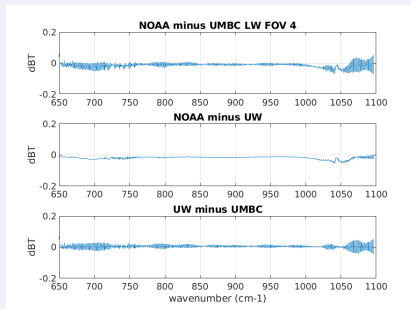
12 Feb 2024

Introduction

- ▶ We make several three-way comparisons of the UMBC, NOAA, and UW/NASA CrIS (Cross-track Infrared Sounder) calibrated radiance products, primarily for the J1/NOAA-20 sounder.
- ▶ The results here are only spot checks of a few granules and FOVs, intended to look for obvious differences or problems. They were chosen as being representative of a much larger number of tests.
- ▶ Overall, the three products look similar, at least for apodized radiances. This is reassuring. Differences in algorithms, and the presence or absence of a polarization correction, do show up as measurable differences.
- ▶ Significant high frequency differences in the unapodized radiances are due in part to UMBC using reference truth with a sinc basis, while UW and NOAA used reference truth convolved with instrument responsivity.
- ▶ One problem with convolving reference truth with responsivity is that the responsivity varies significantly from instrument to instrument. So there is no one reference truth.

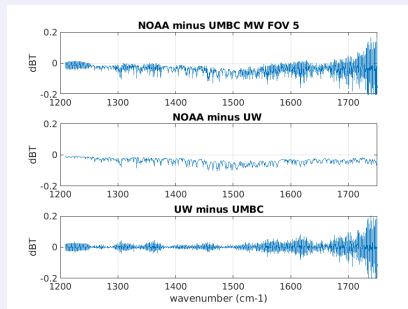
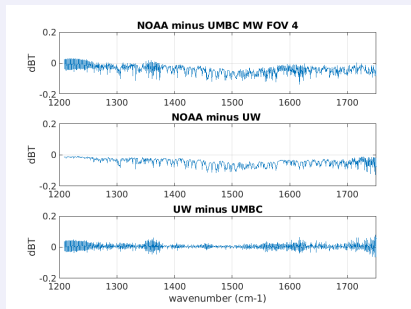
- ▶ UMBC radiances are from the UMBC CCAST versions V20 or V21, c. 2023, using algorithm “c7”, except as noted.
- ▶ UW/NASA radiances are from their L1b V2 product, except as noted.
- ▶ NOAA radiances are from the NOAA CrIS SDR production system, as of late 2023 and early 2024.
- ▶ the CCAST algorithm c7 uses a “ratio-first” calibration algorithm [2], periodic sinc at the sensor grid, cosine apodization at the band edges of the sensor grid, and resampling to take the sensor to the user grid.
- ▶ NOAA and UW use an “SA-first” algorithm [3], a sinc basis with an extended SA matrix, circular shift of the extended user grid, and resampling. NOAA and UW V3 include a polarization correction, while UMBC and UW V2 do not.

LW unapodized comparisons (Oct 14)



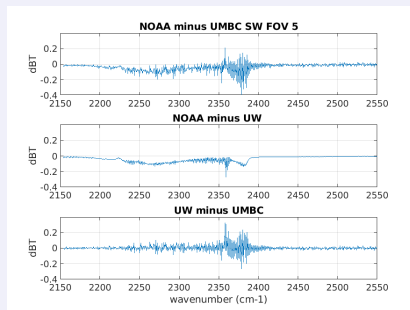
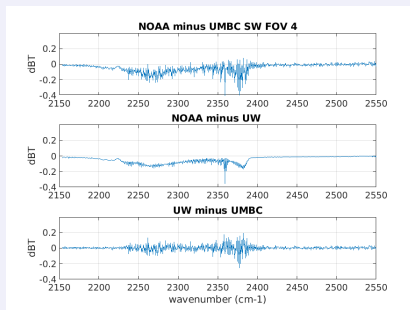
A 3-way comparison of NOAA, UMBC, and UW, LW unapodized radiances, for FOVs 4 and 5. Results are for J1/NOAA-20, 5 Oct 2023, UTC 19:03, lat 55 lon -97, 1200 matching obs, all FORs.

MW unapodized comparisons (Oct 14)



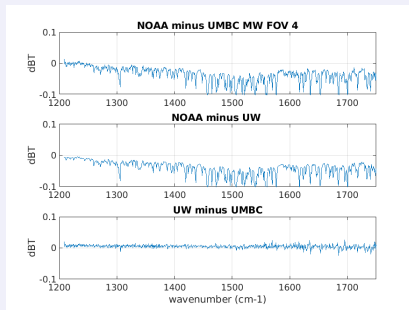
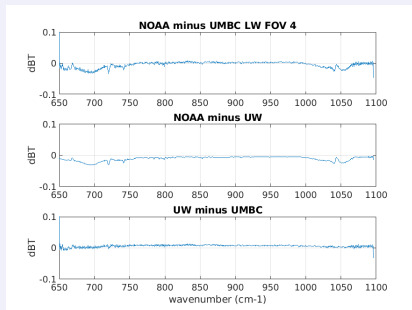
A 3-way comparison of NOAA, UMBC, and UW, MW unapodized radiances, for FOVs 4 and 5. Results are for J1/NOAA-20, 5 Oct 2023, UTC 19:03, lat 55 lon -97, 1200 matching obs, all FORs.

SW unapodized comparisons (Oct 14)



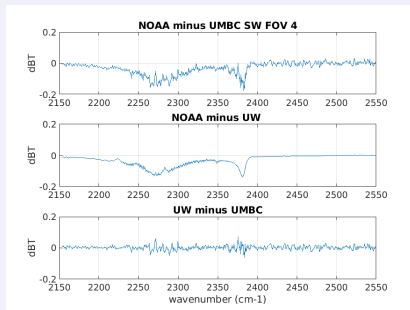
A 3-way comparison of NOAA, UMBC, and UW, SW unapodized radiances, for FOVs 4 and 5. Results are for J1/NOAA-20, 5 Oct 2023, UTC 19:03, lat 55 lon -97, 1200 matching obs, all FORs. Note the range of the Y axis has been extended to ± 0.4 .

LW & MW apodized comparisons (Jan 31)



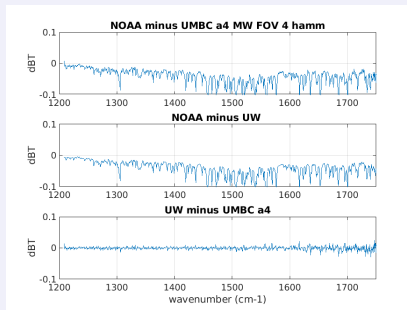
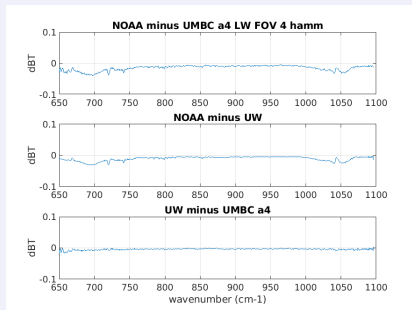
A 3-way comparison of NOAA, UMBC, and UW, LW and MW Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis has been reduced to ± 0.1 .

SW apodized comparisons (Jan 31)



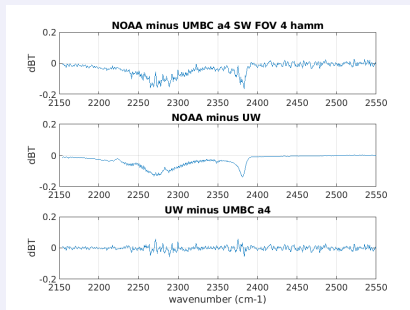
A 3-way comparison of NOAA, UMBC, and UW, SW Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.2K$ for this plot.

apodized a4 comparisons (2 Feb)



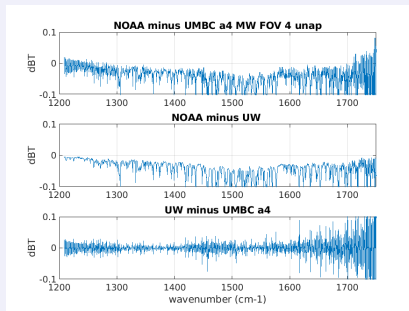
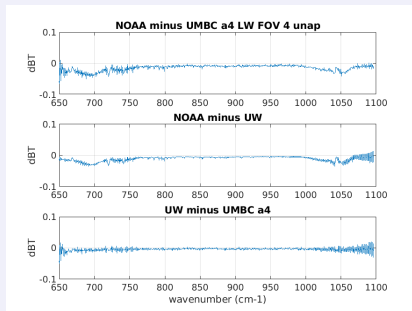
3-way comparison of NOAA, UMBC A4, and UW, LW and MW, Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.1K$ for these plots.

apodized a4 comparisons (2 Feb)



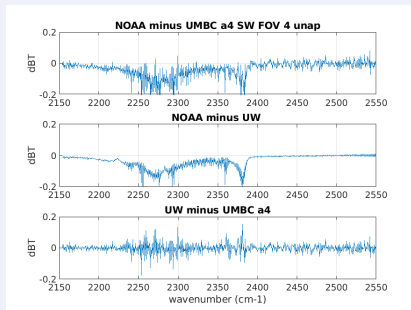
3-way comparison of NOAA, UMBC A4, and UW, SW Hamming apodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.2K$ for these plots.

unapodized a4 comparisons (2 Feb)



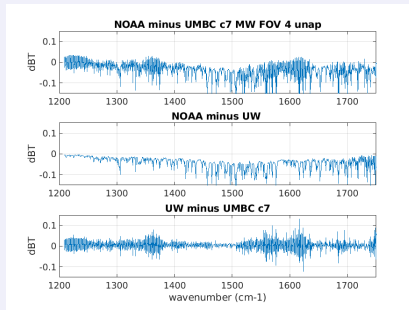
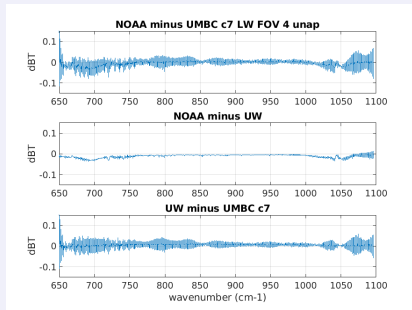
3-way comparison of NOAA, UMBC A4, and UW, LW and MW unapodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.1K$ for these plots.

unapodized a4 comparisons (2 Feb)



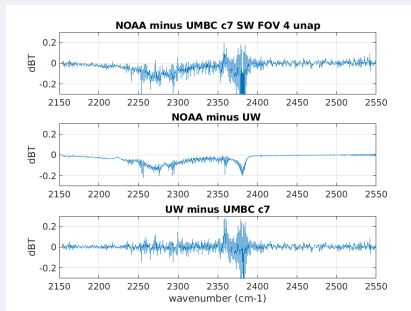
3-way comparison of NOAA, UMBC A4, and UW SW Hamming calibrated radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note the range of the Y axis is $\pm 0.2K$ for these plots.

unapodized c7 comparisons (2 Feb)



3-way comparison of NOAA, UMBC c7, and UW, LW and MW unapodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note this is the same test method as the 5 Oct 2023 UTC 19:03 tests, but with different data. The range of the Y axis is $\pm 0.15K$ for these plots.

unapodized c7 comparisons (2 Feb)

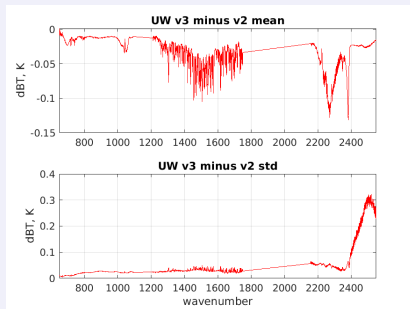


3-way comparison of NOAA, UMBC c7, and UW, SW unapodized radiances, for FOV 4. Results are for J1/NOAA-20, 05-Oct-2023 UTC 07:03:14, lat 21.9 lon 91.8, with 1170 matching obs, all FORs. Note this is the same test method as the 5 Oct 2023 UTC 19:03 tests, but with different data. The range of the Y axis is $\pm 0.3K$ for this plot.

Discussion II

- ▶ This group of plots starts with comparisons in which UMBC is using an "SA first" algorithm (UMBC A4), similar to NOAA and UW, with Hamming apodization applied to all obs. As we might expect, the differences between UMBC A1 and UW are the smallest we have shown. Note the Y axis is $\pm 0.1K$ for the LW and MW plots, and $\pm 0.2K$ for the SW.
- ▶ The corresponding unapodized residuals are also generally smaller than the residuals we showed using UMBC C7.
- ▶ While this agreement is reassuring in some ways, we should remember that UW, NOAA, and UMBC are simply all wrong together here, in using an algorithm validated with variable reference truth.
- ▶ The comparisons using UMBC C7, algorithm are same test method as the 5 Oct 2023 UTC 19:03 tests, but with different data. The range of the Y axis is $\pm 0.15K$ for the LW and MW, and $\pm 0.3K$ for the SW plots.

UW V3 minus V2 (9 feb)



UW V3 minus V2 (9 feb)

Conclusions

- ▶ The met laser residuals are relatively consistent across tests. The default focal plane is not a perfect fit, but has served as a good starting point.
- ▶ In the absolute tabulation we see the FOV 5 residuals are all positive and within a few ppm across tests. This suggests a Neon adjustment.
- ▶ For the J3 tests, using calibrated radiances reduced $\text{rms}(\text{obs} - \text{calc})$, but gave at most very small changes in the met laser residuals, never more than one discretization step, 0.64 ppm for those tests.
- ▶ The reduced $\text{rms}(\text{obs} - \text{calc})$ working from calibrated radiances makes a case for switching to the latter to increase confidence, particularly for the final round of tests.